

Storage in the Cloud TASK – 2

- 1. Create a new storage bucket in your chosen cloud provider (AWS S3, Google Cloud Storage, or Azure Blob Storage) named 'my-music-uploads' and upload any 2 MP3 files to it.**

Ans :

- Create a new S3 bucket
- Sign in to the AWS Management Console.
- Open Amazon S3 from the Services menu.
- Click Create bucket.
- Enter the bucket name:
- my-music-uploads
- Select your preferred region (for example, Asia Pacific (Mumbai) – ap-south-1).
- Keep Block all public access enabled for security.
- Click Create bucket.
- Upload two MP3 files
- Open the newly created bucket: my-music-uploads.
- Click Upload.
- Verify the upload
- Open the bucket's Objects tab.
- Confirm that both MP3 files are listed:
 - favorite-song-1.mp3
 - favorite-song-2.mp3

- 2. Apply a bucket policy to 'my-music-uploads' so only users with a specific email address (your own or a friend's) can download files from the bucket.

Hint:**

Use the bucket's access management settings to add an IAM user or assign a permission for a specific email.

Ans :

- Identify the user who should get access
- Use an AWS IAM user or AWS account associated with the required email address.
- Example email: friend@example.com (replace with your own or your friend's AWS account email).
- Open bucket permissions
- Go to the Amazon S3 Console.
- Open the bucket:
- my-music-uploads
- Add a bucket policy
- Scroll to Bucket policy and click Edit.
- Save the policy
- Click Save changes.
- Ensure Block Public Access remains enabled so files are not available to everyone.
- Test access
- Log in as the authorized IAM user and try downloading an MP3 file.
- Confirm that the permitted user can access the files while unauthorized users cannot.

3. Enable server-side encryption for all objects uploaded to 'my-music-uploads', ensuring that future uploads are encrypted automatically by the cloud provider.

Ans :

- Open your S3 bucket
- Sign in to the AWS Management Console.
- Go to Amazon S3.
- Open the bucket:
- my-music-uploads
- Enable default encryption
- Select the Properties tab.
- Scroll to Default encryption.
- Click Edit.
- Select Enable server-side encryption.
- Choose an encryption type:
 - SSE-S3 (Amazon S3 managed keys) – simple automatic encryption managed by AWS.
 - *(Optional)* SSE-KMS – uses AWS Key Management Service for additional key control.
- Save the encryption settings
- Select SSE-S3 as the default encryption method.
- Click Save changes.
- Verify encryption
- Upload a new MP3 file to the bucket.
- Open the file details.
- Check the Server-side encryption field to confirm that encryption is enabled.

4. Test your bucket's security by trying to access an uploaded MP3 file from a browser where you are not logged in — take a screenshot of the error or access denied message.

Ans :

- Copy the MP3 file URL
- Open the Amazon S3 Console.
- Go to the bucket:
- my-music-uploads
- ☐ Select one uploaded MP3 file.
- Copy the Object URL.
- Open a private browser window
- Open a new Incognito/Private browsing window.
- Make sure you are not signed in to your AWS account.
- Test the file access
- Paste the copied S3 Object URL into the browser.
- Press Enter.

**5. Use an AI tool like ChatGPT or Copilot to generate a sample bucket policy that restricts access to your bucket by IP address, then modify and apply it to 'my-music-uploads'.

Hint: Copy the AI-generated policy, review it for your cloud provider's syntax, and adjust the IP to your college's or your own network.**

Ans :

I used an AI tool to generate a sample AWS S3 bucket policy that restricts access based on an IP address. After reviewing the policy

syntax, I modified the IP address to match my allowed network and applied it to the my-music-uploads bucket.

Sample AWS S3 Bucket Policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "RestrictAccessByIPAddress",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::my-music-uploads/*",
      "Condition": {
        "IpAddress": {
          "aws:SourceIp": "YOUR_IP_ADDRESS/32"
        }
      }
    }
  ]
}
```

I replaced YOUR_IP_ADDRESS/32 with my own network IP address and added the policy under the Permissions → Bucket Policy section of the S3 bucket. After applying the policy, only requests coming from the allowed IP address can access the MP3 files, while requests from other networks are denied.

This provides an additional security layer by limiting bucket access to a trusted network location.